

Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions

Abstract

Large Language Models (LLMs) are increasingly used in research activities including literature synthesis, evidence extraction, hypothesis generation, qualitative coding, and the drafting of research conclusions. However, the reliability of these applications depends not only on model capability but also on the formulation of the prompt. Prompt sensitivity refers to changes in model behavior caused by variations in prompt wording, structure, examples, ordering, or formatting that preserve the intended task. While prior research has established prompt sensitivity as a significant property of modern language models, less attention has been given to its implications for the stability of research conclusions. This distinction is important because a small change in an instruction may alter an answer, but a change in a research conclusion can affect the interpretation of evidence, theoretical claims, or practical recommendations.

This paper synthesizes scholarly research on prompt sensitivity, in-context learning, prompting strategies, LLM factuality, reproducibility, and output consistency to examine how prompt variation can propagate through a research workflow. The paper argues that conclusion instability can arise through several mechanisms: lexical and semantic prompt variation, few-shot example selection and ordering, instruction framing, model stochasticity, evaluation artifacts, and the interaction between prompting and evidence supplied to the model. Existing approaches—including prompt calibration, prompt ensembling, chain-of-thought prompting, self-consistency, structured prompting, retrieval augmentation, and sensitivity metrics—can reduce some forms of variability but do not eliminate the fundamental problem.

The paper proposes treating prompt choice as part of the research measurement instrument rather than as an informal interface decision. Future research should therefore evaluate not merely whether an LLM produces a plausible conclusion, but whether the conclusion remains invariant, or at least appropriately uncertain, across a predefined family of semantically equivalent prompts. Such an approach could strengthen reproducibility, methodological transparency, and epistemic reliability in LLM-assisted research.

Keywords: Large Language Models; prompt sensitivity; prompt robustness; research conclusions; reproducibility; LLM reliability; prompt engineering; factuality; uncertainty; scientific research

1. Introduction

Large Language Models (LLMs) have developed from systems primarily evaluated on language-generation benchmarks into general-purpose tools for knowledge-intensive work. Researchers increasingly employ LLMs to summarize literature, classify documents, extract variables, formulate hypotheses, synthesize evidence, generate analytical interpretations, and assist with the writing of scholarly manuscripts. These applications create a methodological question that is distinct from conventional model accuracy: Would the research conclusion remain the same if the researcher expressed the same analytical request using slightly different wording?

This question concerns prompt sensitivity. An LLM does not receive a research task as an abstract logical proposition. It receives a sequence of tokens whose particular lexical, syntactic, structural, and contextual configuration influences the probability distribution over subsequent tokens. Consequently, two prompts that are semantically equivalent to a human researcher can produce different outputs. Early work on few-shot learning demonstrated that prompt format, example selection, and example ordering could cause large differences in performance (Zhao et al., 2021; Lu et al., 2022). More recent work has developed explicit metrics for measuring sensitivity to prompt paraphrases and has shown that sensitivity varies across models, datasets, tasks, and individual instances (Zhuo et al., 2024; Errica et al., 2025).

For ordinary question answering, prompt sensitivity may appear as an inconvenience: one formulation produces a correct answer while another produces an incorrect or less useful answer. In research settings, however, the implications are more serious. If an LLM is used to synthesize a body of literature and its conclusion changes when the prompt is rephrased, then the resulting conclusion is partly dependent on an undocumented linguistic choice. This can undermine reproducibility because another researcher following the same conceptual procedure may obtain a different result even when using the same source material and model.

The problem is amplified when LLM output becomes an intermediate variable in a larger research pipeline. A prompt may first determine whether studies are considered relevant; the selected studies may then be summarized; extracted evidence may subsequently be synthesized; and the synthesis may finally be converted into a conclusion. Prompt-induced differences at an early stage can therefore propagate through subsequent stages. Recent research on systematic-review screening illustrates this concern: model performance can change substantially according to how eligibility criteria and task instructions are formulated (Strachan, 2024).

This paper examines the relationship between prompt sensitivity and the stability of LLM-generated research conclusions. Its central argument is that conclusion stability should be treated as a separate methodological property from ordinary task accuracy. A model may produce an accurate conclusion under one prompt while remaining highly

sensitive to semantically equivalent alternatives. Conversely, a model may exhibit linguistic variation while preserving the underlying conclusion. Therefore, evaluation should distinguish superficial output variation from epistemically meaningful conclusion changes.

The paper addresses five questions:

- 1. What is prompt sensitivity and what mechanisms contribute to it?**
- 2. How can prompt variation affect research-oriented LLM outputs?**
- 3. What existing techniques attempt to improve prompt robustness and consistency?**
- 4. What methodological limitations prevent current approaches from guaranteeing stable conclusions?**
- 5. What research agenda could establish reproducible standards for evaluating LLM-generated conclusions?**

2. Background and Related Work

2.1 Large Language Models as Conditional Generative Systems

At a high level, an autoregressive LLM estimates a probability distribution over a sequence of tokens:

$$P(y|x) = \prod_{t=1}^T P(y_t | x, y_{<t}),$$

where x represents the prompt and y represents the generated output. A prompt is therefore not merely an instruction in a symbolic programming sense; it forms part of the statistical context conditioning generation.

This distinction helps explain why meaning-preserving textual changes can influence output. Suppose x_1 and x_2 are two prompts that communicate essentially the same research task. Ideally,

$$f(x_1, D) \approx f(x_2, D),$$

where D is the evidence supplied to the model and f is the model's inference process. In practice, the generated responses can differ because the model processes the exact token sequence rather than a canonical semantic representation.

Research on in-context learning provided some of the earliest systematic evidence. Zhao et al. (2021) found that GPT-3 performance could vary substantially with prompt format, selected examples, and example order. Their contextual-calibration method reduced some of this instability by correcting biases in model predictions.

Similarly, Lu et al. (2022) demonstrated that merely changing the ordering of few-shot examples could move performance from near state-of-the-art levels toward random-guess performance. Importantly, the best ordering was not necessarily transferable between models, showing that prompt sensitivity interacts with model-specific behavior.

These findings establish that prompt design is not an incidental interface detail. It can function as a substantive experimental variable.

2.2 From Prompt Sensitivity to Prompt Robustness

Prompt sensitivity can be conceptualized as the degree to which model behavior changes under meaning-preserving perturbations of the input. A simple conceptual measure is:

$$S(x) = E_{x' \sim P(x)}[d(f(x), f(x'))],$$

where $P(x)$ is a set of acceptable prompt variants and d is a distance between outputs.

The distance function is critical. Lexical distance alone is insufficient because two outputs can be textually different but convey the same conclusion. For research applications, a more useful distinction is between:

- surface instability: differences in wording, organization, or explanation;
- claim instability: differences in factual or analytical claims;
- conclusion instability: differences in the final interpretation or answer to the research question.

Zhuo et al. (2024) introduced ProSA and the PromptSensiScore to investigate prompt sensitivity at the instance level. Their experiments found that sensitivity differs across datasets and models, while larger models tended to show greater robustness. They also found that few-shot examples could reduce sensitivity and that subjective evaluations themselves can be prompt-sensitive, particularly for complex reasoning tasks.

Errica et al. (2025) further distinguished sensitivity from consistency. Their sensitivity measure captures prediction changes across prompt rephrasings without requiring ground-truth labels, whereas consistency characterizes variation across rephrasings for items belonging to the same class.

These distinctions are directly relevant to research conclusions. A methodology that only measures answer accuracy may miss an important failure mode: an LLM can be correct for the evaluated prompt while remaining unstable across plausible alternative formulations.

2.3 Prompting and Reasoning

Prompt engineering has developed into a broad collection of techniques, including zero-shot prompting, few-shot prompting, chain-of-thought (CoT) prompting, self-consistency, role prompting, decomposition, structured outputs, and retrieval-based approaches. The Prompt Report by Schulhoff et al. (2024) systematizes this rapidly expanding literature and identifies a large taxonomy of prompting techniques.

Chain-of-thought prompting is particularly important for research-oriented tasks because it attempts to elicit intermediate reasoning steps. Wei et al. (2022) demonstrated substantial gains on arithmetic, commonsense, and symbolic reasoning tasks when models were provided with reasoning demonstrations. Kojima et al. (2022) subsequently showed that a simple zero-shot instruction could improve reasoning performance on several benchmarks.

However, improved reasoning performance should not automatically be equated with stable research conclusions. If a reasoning instruction changes the path through which a model reaches a conclusion, it can also change the conclusion itself. Reasoning prompts therefore represent both a possible robustness intervention and an additional source of prompt-dependent variation.

3. Prompt Sensitivity and the Stability of Research Conclusions

3.1 Why Research Conclusions Are a Special Case

A research conclusion is not simply an answer string. It represents an interpretation of evidence under specified assumptions. In a conventional empirical study, the conclusion is intended to follow from a transparent chain connecting data, methods, analysis, and inference.

An LLM-assisted workflow may instead have the form:

$$D \rightarrow P \rightarrow M \rightarrow O \rightarrow C$$

where:

- D = research evidence or dataset,
- P = prompt,
- M = model,
- O = generated output,
- C = extracted research conclusion.

If P changes while D and M remain fixed, any change in C is attributable, at least partly, to prompt variation. This makes prompt selection analogous to a methodological choice.

The problem is particularly important when prompts instruct models to synthesize conflicting evidence. Consider two semantically equivalent requests:

“Summarize whether the evidence supports the hypothesis.”

and

“Assess the extent to which the available evidence provides support for the hypothesis.”

A human researcher would normally expect approximately the same interpretation. Yet a generative model may produce different emphases, evidence selections, confidence levels, or final judgments.

The difference becomes more consequential when prompts contain evaluative framing. Instructions such as “identify evidence supporting the hypothesis” and “critically assess whether the evidence supports the hypothesis” may encourage different evidence-selection behavior even when both concern the same research question.

3.2 Sources of Prompt-Induced Instability

Prompt sensitivity can originate from several dimensions.

Lexical variation

Replacing words with synonyms, changing sentence structure, or modifying politeness and directness can affect generation. Recent research specifically characterizes prompt sensitivity as a response to surface-level or meaning-preserving changes.

Structural variation

The same information can be arranged as prose, bullet points, a numbered procedure, or a structured schema. Structural changes can modify which portions of the prompt receive greater contextual influence.

Demonstration selection and ordering

Few-shot examples provide implicit task definitions. Zhao et al. (2021) and Lu et al. (2022) show that changing examples or their order can produce substantial performance differences.

Instruction framing

An instruction emphasizing skepticism, completeness, concision, or agreement can change the response distribution. For research synthesis, framing can influence which evidence is emphasized and how uncertainty is expressed.

Output constraints

Requests such as “provide a definitive conclusion” versus “report the strength of evidence and remaining uncertainty” can alter the epistemic character of the generated answer.

Decoding stochasticity

Prompt sensitivity is not the same as output randomness. Even with an identical prompt, stochastic decoding can produce different responses. Self-consistency methods exploit this fact by generating multiple reasoning paths and aggregating them (Wang et al., 2023).

Consequently, robust evaluation should consider both within-prompt variability and between-prompt variability.

4. Mechanisms Through Which Prompt Sensitivity Can Affect Conclusions

4.1 Evidence Selection

An LLM synthesizing literature may not treat every piece of evidence equally. Prompt wording can influence which findings are foregrounded, summarized, or interpreted as central.

For example, a prompt emphasizing “positive findings” can encourage the model to prioritize supportive evidence, whereas a prompt emphasizing “limitations and contradictory evidence” may produce a more cautious synthesis. The issue is not merely stylistic: different evidence-selection patterns can produce different conclusions.

This creates an important distinction between retrieval stability and interpretation stability. A research system may retrieve the same documents under multiple prompts while interpreting those documents differently.

4.2 Reasoning-Path Variation

CoT prompting demonstrates that explicit reasoning instructions can change the way models approach complex tasks (Wei et al., 2022). Self-consistency extends this idea by sampling multiple reasoning paths and selecting an answer supported by several paths (Wang et al., 2023).

For research synthesis, multiple reasoning paths may reveal whether a conclusion is robust. If ten independent reasoning paths converge on the same conclusion, this provides some evidence of stability. If they divide between competing interpretations, the disagreement itself is valuable information.

However, convergence is not equivalent to truth. Multiple generations may share the same underlying model biases or erroneous assumptions. Thus, consensus among generations should not be interpreted as independent confirmation.

4.3 Confidence and Factuality

Prompt sensitivity interacts with the broader problem of LLM factuality. TruthfulQA demonstrated that language models can produce plausible but false statements, including responses reflecting common misconceptions (Lin et al., 2022).

Likewise, Lee et al. (2022) investigated factuality in open-ended generation and found that generation methods can affect factual accuracy. Their work illustrates why a fluent and internally coherent conclusion cannot automatically be treated as evidence-supported.

Prompt sensitivity can therefore affect both what conclusion is generated and how confidently it is expressed. A particularly problematic case occurs when one prompt causes an uncertain conclusion while another produces an apparently definitive conclusion despite identical evidence.

4.4 Propagation Through Research Pipelines

Prompt-induced differences can accumulate across sequential tasks:

$$D \rightarrow R1 \rightarrow S1 \rightarrow R2 \rightarrow C$$

where R_i represents intermediate reasoning or extraction and S_i represents selected evidence.

Suppose an LLM first screens 1,000 papers. A small prompt change alters which studies are included. A second LLM then summarizes the selected studies, and a third stage synthesizes the summaries. Even if each individual stage has only moderate sensitivity, the final conclusion can diverge substantially.

Systematic-review research demonstrates why this matters. Strachan (2024) reported that prompts designed for article screening were strongly dependent on the exact instructions and used iterative prompt development to obtain robust sensitivity across alternative eligibility-criteria formulations.

Thus, prompt sensitivity should be considered a pipeline-level phenomenon, not merely an output-level phenomenon.

5. Current Approaches and Developments

5.1 Prompt Calibration

Contextual calibration attempts to correct systematic biases introduced by the prompt. Zhao et al. (2021) demonstrated that calibration can improve accuracy while reducing variance across prompt choices.

For research conclusions, calibration could be interpreted as an attempt to ensure that irrelevant prompt features exert less influence on the final answer. Its limitation is that calibration is primarily effective for particular output distributions and task types. It does not establish that an open-ended scientific conclusion is invariant under semantic prompt perturbations.

5.2 Prompt Ensembling

Rather than selecting one prompt, researchers can construct a family of prompts:

$$P = \{P_1, P_2, \dots, P_n\}$$

and aggregate the resulting conclusions. This reduces dependence on a single formulation.

For classification, majority voting is straightforward. For open-ended research conclusions, aggregation is harder because two responses may use different language while expressing the same claim.

A practical solution is to transform each response into structured claims and compare them using semantic similarity, entailment, or human evaluation. The goal is not necessarily identical wording but stable claim-level content.

5.3 Self-Consistency

Self-consistency samples multiple reasoning paths from a single prompt and selects an answer that is most consistently supported. Wang et al. (2023) showed strong improvements across several reasoning benchmarks.

For research applications, self-consistency can be extended from a single prompt to multiple prompts. Instead of:

$$P \rightarrow \{O_1, O_2, \dots, O_n\},$$

a more robust procedure is:

$$\{P_1, P_2, \dots, P_k\} \rightarrow \{O_{ij}\} \rightarrow \text{claim aggregation.}$$

This creates a two-dimensional robustness test involving both prompt variation and generation variation.

5.4 Structured Prompting

Structured prompts explicitly separate:

- 1. research question;**
- 2. evidence;**
- 3. analytical criteria;**
- 4. assumptions;**
- 5. limitations;**
- 6. output requirements.**

This can reduce ambiguity and make the task more reproducible. Structured prompting is particularly useful because it distinguishes what evidence is available from what conclusion is requested.

However, structure itself may become a sensitivity variable. A highly elaborate template may perform differently from a concise template even when both encode the same task. Therefore, structured prompting should be evaluated rather than assumed to be inherently robust.

5.5 Retrieval-Augmented and Evidence-Grounded Generation

Retrieval-augmented generation (RAG) can constrain LLM outputs by providing explicit evidence. This is particularly valuable for scholarly research because it reduces reliance on unsupported model memory.

Nevertheless, RAG does not automatically solve prompt sensitivity. The prompt can influence:

- which retrieved documents are considered relevant;
- how evidence is summarized;
- how contradictory findings are reconciled;
- which sources are cited;
- the final interpretation.

Consequently, evidence grounding should be combined with prompt-sensitivity testing rather than treated as a complete solution.

5.6 Sensitivity and Consistency Metrics

Recent research has increasingly moved from qualitative observations toward explicit metrics. ProSA introduced PromptSensiScore, while Errica et al. (2025) proposed complementary sensitivity and consistency measures.

This represents an important development because robustness can now become an empirical property of an LLM system rather than an informal judgment.

For research conclusions, however, metrics must operate at the claim level. A difference in sentence length should not be treated as equivalent to a reversal of the conclusion.

6. Research Challenges and Limitations

6.1 Defining Semantic Equivalence

The most fundamental challenge is determining whether two prompts actually express the same task.

Human judgments of semantic equivalence can be subjective. Automated semantic similarity measures can also fail when two prompts use similar words but impose different analytical constraints.

A robust benchmark therefore requires carefully constructed perturbation sets containing categories such as:

- synonym replacement;
- grammatical restructuring;
- active/passive transformation;
- changes in politeness;
- formatting changes;
- ordering changes;
- instruction paraphrases;
- irrelevant contextual additions.

6.2 Distinguishing Output Variation from Conclusion Variation

Two outputs may differ considerably while agreeing on the core conclusion. Conversely, two nearly identical outputs may differ on one critical causal or factual claim.

Therefore, stability should be measured at multiple levels:

Level | Example | Stability criterion

Lexical | Different wording | Low importance

Semantic | Different explanation | Moderate importance

Claim | Different factual assertion | High importance

Conclusion | Different answer to research question | Critical

Recommendation | Different practical implication | Critical

This hierarchy is necessary for meaningful research evaluation.

6.3 Model and Version Dependence

Prompt sensitivity is model-dependent. Zhuo et al. (2024) found differences across models and datasets, while Lu et al. (2022) showed that good prompt orderings need not transfer between models.

Consequently, a prompt validated for one model cannot automatically be assumed to be robust for another. Model updates create an additional reproducibility problem: the same prompt may behave differently after a provider changes the underlying model.

6.4 Evaluation Artifacts

Prompt sensitivity may sometimes be exaggerated by the evaluation procedure itself. Hua et al. (2025) specifically revisited prompt sensitivity and argued that some observed sensitivity can arise from evaluation methods such as rigid answer matching and likelihood-based scoring. Their experiments suggest that semantically correct responses may be incorrectly treated as different when evaluation is overly surface-oriented.

This finding is particularly important for research conclusions because evaluation must distinguish genuine epistemic disagreement from stylistic variation.

6.5 Lack of Ground Truth for Complex Research Questions

Many research questions do not have an immediately available ground-truth answer. If an LLM produces two competing interpretations of a literature base, an evaluator cannot always determine which one is objectively correct.

This creates a distinction between:

- accuracy, which requires an appropriate reference answer;
- consistency, which can be evaluated without knowing the correct answer;

- evidence faithfulness, which evaluates whether claims are supported by supplied sources.

Research-oriented LLM evaluation therefore requires multiple complementary criteria.

6.6 Reproducibility and Documentation

Scientific reproducibility requires sufficient information to reconstruct an analysis. NLP research has increasingly recognized this issue. Magnusson et al. (2023), analyzing more than 10,000 reproducibility-checklist responses, found that reporting practices influence the reproducibility of NLP research and emphasized the importance of documenting experimental details.

For LLM research, documentation should include not only the model name but also:

- exact prompt;
- prompt version;
- system instructions where applicable;
- model/version identifier;
- decoding parameters;
- temperature and sampling configuration;
- retrieval corpus;
- retrieved documents;
- demonstrations;
- output schema;
- number of repeated generations;
- evaluation procedure.

Without this information, prompt-sensitive results may be impossible to reproduce.

7. Research Gaps and Future Directions

7.1 From Prompt Robustness to Conclusion Robustness

Existing prompt-sensitivity research often evaluates classification accuracy or response similarity. A major research gap is the systematic measurement of research-conclusion stability.

Future benchmarks should provide a research question, evidence corpus, and a family of semantically equivalent prompts. The system should then generate conclusions for every prompt. Evaluation should determine:

$$CS = 1 - N_{\text{conclusion changes}} / N_{\text{valid prompt comparisons}},$$

where CS denotes conclusion stability.

This metric should be supplemented by claim-level analysis so that a minor explanatory change is not treated as a conclusion reversal.

7.2 Prompt Families as Experimental Units

A major conceptual improvement would be to treat the prompt family rather than the individual prompt as the experimental unit.

Instead of reporting:

“The model concludes X.”

researchers should report:

“Across 50 meaning-preserving prompt variants, the model reached conclusion X in 46 cases and an alternative conclusion in 4 cases.”

Such reporting would make prompt-induced uncertainty visible.

7.3 Prompt Uncertainty as Measurement Uncertainty

Prompt choice can be understood as a source of measurement uncertainty. Recent work has explicitly proposed the concept of prompt uncertainty, treating differences caused solely by prompt wording as variation in the measurement process (Leng, 2026).

This perspective is promising because it connects prompt sensitivity with established scientific ideas such as measurement error, robustness analysis, sensitivity analysis, and uncertainty quantification.

A research result could therefore be expressed as:

$$\hat{C} = C(P, D, M),$$

with uncertainty arising from the admissible prompt family:

$$UP = \text{Var}_{P \in \mathcal{P}} [C(P, D, M)].$$

A low UP indicates that the conclusion is relatively insensitive to prompt choice; a high UP indicates that the conclusion depends strongly on linguistic formulation.

7.4 Adversarial and Naturalistic Prompt Perturbations

Future studies should distinguish between artificially constructed perturbations and realistic researcher behavior. Researchers rarely change only one synonym; they modify prompts by adding context, reorganizing instructions, inserting examples, changing output requirements, or asking follow-up questions.

Consequently, robustness benchmarks should contain both:

- controlled minimal perturbations; and
- realistic prompt variants generated independently by human researchers.

7.5 Human–LLM Collaborative Validation

The objective should not be to make LLM conclusions perfectly invariant. Some prompt changes legitimately should change a conclusion because they introduce different evidence, assumptions, or analytical criteria.

The research goal should instead be appropriate sensitivity:

the model should be insensitive to irrelevant changes and sensitive to relevant changes.

This distinction is crucial. A model that gives exactly the same conclusion regardless of meaningful changes in evidence is not robust—it is unresponsive.

Future evaluation should therefore test two complementary properties:

Robustness = Invariance to irrelevant changes + Responsiveness to relevant changes.

7.6 Standardized Reporting Protocols

A standardized protocol could require researchers using LLMs for evidence synthesis to report:

1. the primary prompt;
2. the admissible prompt family;
3. the number and types of prompt perturbations;
4. model/version and decoding settings;
5. output consistency;
6. claim-level disagreements;
7. evidence citations supporting each conclusion;

8. human adjudication of disputed conclusions.

Such reporting would transform prompt engineering from an informal craft into a reproducible methodological practice.

8. Conclusion

Prompt sensitivity represents an important but underappreciated threat to the stability of LLM-generated research conclusions. Evidence from few-shot learning, prompt-order studies, dedicated sensitivity metrics, and recent robustness research demonstrates that seemingly minor prompt changes can alter model behavior (Zhao et al., 2021; Lu et al., 2022; Zhuo et al., 2024; Errica et al., 2025).

For research applications, the problem extends beyond ordinary answer variability. Prompt changes can affect evidence selection, reasoning paths, confidence, factual claims, and ultimately the conclusion presented to the researcher. When LLMs are integrated into multi-stage workflows, these differences may propagate through the pipeline and produce substantially different research outcomes.

Existing techniques—including contextual calibration, prompt ensembling, structured prompting, chain-of-thought reasoning, self-consistency, retrieval augmentation, and sensitivity metrics—provide useful partial solutions. However, none establishes that a conclusion is scientifically reliable merely because it is generated consistently under one carefully designed prompt. Likewise, agreement among multiple model outputs cannot substitute for evidence-based validation.

The most important methodological shift is therefore to regard prompt choice as part of the measurement process. Research employing LLMs should evaluate whether conclusions remain stable across a predefined set of semantically equivalent prompts while also testing whether the model responds appropriately when evidence or analytical assumptions genuinely change.

Future research should move from measuring prompt performance toward measuring prompt uncertainty, claim stability, and conclusion robustness. A mature standard for LLM-assisted research should report not only the conclusion generated by a model, but also the range of conclusions produced under reasonable alternative prompt formulations.

Ultimately, the objective is not to eliminate prompt sensitivity entirely. Rather, reliable LLM-assisted research requires distinguishing irrelevant linguistic sensitivity from legitimate analytical sensitivity. Establishing this distinction is essential if LLMs are to function as reproducible instruments in scholarly research rather than as opaque generators whose conclusions depend unpredictably on the wording of an instruction.

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